Assignment 1

Instructions:

- **Due: 26 Sep, 23:59** (Canvas)
- **Submit**: One MS Word file named MGS3101_A1_YourName

Question 1:

A survey handed out to individuals at a major shopping mall in the Mixc (Wenzhou) in Sep 2025 asked the following:

- Gender
- Age
- Ethnicity (Chinese, Italian, Indian, Eurasian, others)
- Length of residency in Wenzhou
- Overall satisfaction with shopping mall services (using a scale of 1-5, going from 1 being poor to 5 being excellent)
- Quality of food (using a scale of 1-5, going from 1 being poor to 5 being excellent)

(a) What types of data (categorical, ordinal, interval, or ratio) would each of the survey items represent and **why**?

(b) Why is it important to define the data type in your data file correctly before conducting data analyses? Use one or two examples to explain your answer.

Question 2:

A university's Online Learning Portal Department (OLPD) collect clickstream data from students using Canvas (e.g., pages viewed, courses attended, links clicked, files upload/download, and timestamps of these activities). With these data, the OLPD conducts analytics to improve its services for the students.

(a) For each type of analytics introduced in Week 1 – descriptive, predictive, prescriptive – provide an example of a managerial decision that could be made using the clickstream data available. In your answer:
   - Identify the decision
   - Specify the metrics involved
   - Explain the value created for students and the university

(b) The OLPD also want to evaluate the usefulness of Canvas. In particular, it would like to evaluate the impact of Canvas on learning outcomes (e.g. job placements, starting salaries) and how the student demographics (e.g. age, gender, family income levels, marital status, etc.) affect their online learning behavior and learning outcomes. While the OLPD has the data from the online learning portal system, it does not have the data on student demographics, job placements and their starting salaries. What are the challenges that the department might encounter in getting the data it needs for this analytics project?